

Tomato Disease & Insect Control Manual With Variety Selection

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Suggested Spray Schedules attached at end.

Acknowledgements

Special thanks goes to the following people and sources of information for making this manual and short course possible.

- UT Extension 2011 Commercial Vegetable Disease Control Guide
- Southeastern U.S. 2011 Vegetable Crop Handbook
- Dr. Allen Straw, Virginia Tech Vegetable Specialist
- Cornell University Department of Plant Pathology Website, T. A. Zitter for pictures
- Diane Brickey – Editing

Disease Identification

Pictures taken from Cornell University, Rutgers University, A.F. Sherf, R. Providenti, and Grainger County Producers



Calcium Deficiency or Blossom-End Rot use Calcium Nitrate the 4th, 6th and 9th week of crop at 50 lbs/acre in drip irrigation.



Magnesium Deficiency
use Epsom Salt at 20 lb/A
or 4 lb/greenhouse



Phytophthora Blight



Septoria Leaf Spot





Botrytis Gray Mold



Leaf Mold



Bacterial Pith Necrosis



Bacterial Canker



Bacterial Speck



Bacterial Spot



Southern Bacterial Wilt

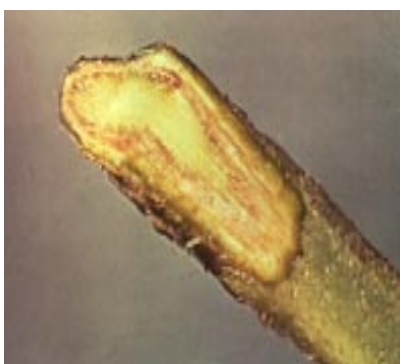




Early Blight



Tomato Spotted Wilt Virus caused by Thripes



Fusarium Wilt



Verticillium Wilt



Late Blight

Virus Diseases and Disorders of Tomato



Tomato Mosaic



Tobacco Mosaic Virus



Cucumber Mosaic



Tomato Big-Bud Phytoplasma

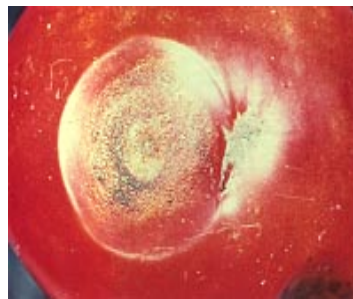
Common Tomato Fruit Disorders



Blotchy Ripening



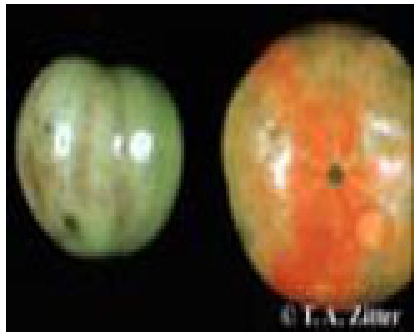
Bacterial Soft Rot



Anthracnose



Graywall



Zippering



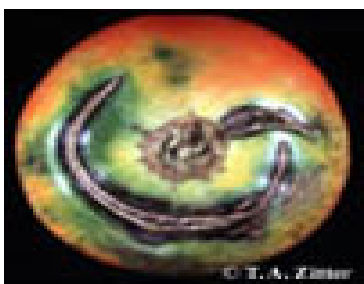
Radial Cracking



Catface



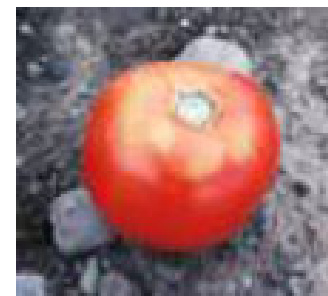
Bird's Eye Spot



Concentric Crack



Sunscauld



Yellow Shoulders

Tomato Disease Control

All tables taken from UT Extension 2011 Commercial Vegetable Disease Guide

Pest	Fungicide	Suggested rate/acre	Remarks & Precautions
TOMATO, PLANT BED, OUTDOOR			
Damping Off Lower stem shrivels and seedling collapses.	Terraclor 75WP	4-8 oz/100 gal water	Soil drench applied to 400-800 sq. ft of containers or beds. Can be repeated once 4-6 weeks later. Use fungicide treated seed when possible.
	Previcur Flex	32 fl oz/1000 sq ft at seeding 16 fl oz after emergence	Minimum of 50 gal water per 1000 sq ft
Botrytis (Gray Mold) Gray, fuzzy growth. & Early Blight Brown spots on leaves or stem.	chlorothalonil 6L	1.38 - 2 pt/43560 sq ft	Spray, first true leaves, repeat at weekly interval. Outdoor beds only.
Bacterial Canker, Spot, Speck Tiny, dark brown to black spots on leaves.	streptomycin sulfate	1 lb/100 gal	When symptoms appear and 4-5 days intervals until transplant
	sodium hypochlorite (Clorox)	1 qt in 4 qt water	Wash seed for 40 min in solution with continuous agitation; air dry promptly. Use 1 gal solution per 1 lb seed

Pest	Fungicide	Suggested Rate/Acre	PHI days	Maximum Use/Acre/Season	Remarks & Precautions
TOMATO, GREENHOUSE After Transplant					
Early & Late Blight, Gray Leaf Spot, Leaf Mold	mancozeb 80WP	1.5 - 2 lb/100 gal	5	22.4 lb	Potential for phytotoxicity(injury) exists for Catamaran. Do not apply as a mixture with any other product. Tanos must be tank mixed with mancozeb.
	Catamaran	4.5 pt/43,560 sq ft	0	50 pt	
	Tanos 50WG	6-8 oz/43,560 sqft	3	72 oz	
Botrytis Stem Canker	Botran 75WP	1 lb/100 gal	10	4 app	Botran is sprayed to stem of plant from ground level up to 18- 24 in
Bacterial Speck & Spot	Kocide DF	2 - 4 Tbsp/1000 sq ft			Foliar sprays
Early & Late Blight, Septoria Leaf Spot	Kocide 2000	1.5 - 2.25 Tbsp/1000 sq ft			
Gray Mold (Botrytis)	Scala 5SC	7 fl oz/100 gal	1	35 fl oz	Foliar sprays. Do not make more than 2 consecutive applications of Degree before rotating. Scala must be tank mixed with another effective fungicide. Ventilate for at least 2 hours after application to avoid plant damage from vapor.
	Catamaran	4.5 pt/43,560 sq ft	0	50 pt	
	Degree 50WDG	1.5 lb/43,560 sq ft	0	6 lb	
	Serenade	2-6 qt/43,560 sqft	0		
Sclerotinia Stem Rot (Timber Rot)	Contans WG	.75 – 1.5 oz/1000 sq ft			Apply to soil 3 months prior to planting. Till 2-8 in depth. Botran should give some control, also.
Powdery Mildew	Rally 40WP	2.5-4 oz/43,560 sq ft			Spray at 1st sign of mildew and repeat at 5- to 14-day intervals. Due to sulfur's high effectiveness extended spray intervals may be possible. Re-apply only if mildew resumes activity. Do not apply if temps will exceed 90F within 3 days after spraying.
	Sulfur 90WP	5 lb/43,560 sq ft			
Pythium Root Rot	Previcur Flex	12.8 fl oz/100 gal	5	4 app	Applied with drip system. See label directions.
	Terramaster4EC	6.5 fl oz/500 gal	3	4 app	

Pest	Fungicide	Suggested Rate/Acre	PHI days	Maximum Use/Acre/Season	Remarks & Precautions
TOMATO, FIELD					
Bacterial Spot and Speck Small, dark spots on foliage and fruit.	Actigard 50WG	0.33 - 0.75 oz	14	8 app	Actigard is a plant resistance activator. Under certain conditions, this product may lead to reductions in yield. Refer to label for disclaimer. Begin applications within 1 week of transplanting. Make up to 8 applications, at weekly intervals. After the eighth application, switch to copper sprays if bacterial diseases are present. Begin Actigard applications at 0.33 oz/A, increasing to 0.75 oz/A as plants grow.
Bacterial Spot, Speck, and Canker Small, dark spots on foliage and fruit.	fixed copper + mancozeb 80WP	See label 1.5 lb	5	22.4 lb	Ridomil/ Copper can be used, but see label for limitations on use of Ridomil. Adding Tanos to copper and mancozeb may slightly enhance efficacy.
	AgriPhage	1 pt			May be useful where copper-resistant bacterial strains are present. Cannot be tank-mixed with copper. Use as part of cooperative program with Omni-Lytics, (866-285-2644), who formulates the bacteriophage to match your bacterial strains. Re-sample frequently, to accommodate strain shifts.
Blossom End Rot Firm, sunken area on blossom end of fruit.	calcium chloride	4 lb/100 gal	0	4 app	Apply as soon as problem is detected or earlier. Maintain adequate calcium level in soil and uniform soil moisture. Avoid excessive irrigation when plants are small.

Pest	Fungicide	Suggested Rate/Acre	PHI day	Maximum Use/Acre/Season	Remarks & Precautions
TOMATO, FIELD					
Early Blight Brown target spots followed by blighting of foliage.	Prior to Fruit Set Cabrio 20EG chlorothalonil 6L mancozeb 80WP Quadris 2.08F Quadris Opti 5.5SC Tanos 50WG	8 - 16 oz 1.38 - 2 pt 1.5 - 3 lb 5 - 6.2 fl oz 1.6 pt		6 app 20 pt 22.4 lb 32 fl oz 5 app	Start spraying soon after plants are set and repeat at 7- to 10-day intervals. Notes for Quadris: Do not apply until 21 days after transplanting. Alternate with a non-related fungicide. Do not mix with an adjuvant. Use higher rate when late blight is present. Notes for Cabrio: See label for precautions on tank mixes. Use higher rate only when late blight is a threat, and alternate with a non-strobilurin fungicide. Notes for late blight: Most strains are resistant to Ridomil. Control of late blight by chlorothalonil, mancozeb, or maneb can be improved by using high pressure sprays and by adding fixed copper. The late blight fungus can overwinter in potato tubers and in greenhouse tomato plants. Notes for Tanos: Must be tank mixed with and alternated with a non-Group 11 fungicide such as chlorothalonil, mancozeb, or copper.
Late Blight (also see late blight, below) Dark, irregular spots, often worse on younger leaves. Favored by mild, wet weather.	After Fruit Set Cabrio 20EG chlorothalonil 6L mancozeb 80WP Quadris 2.08F Quadris Opti 5.5SC Tanos 50WG	6 - 8 oz		72 oz	
Anthracnose Circular, sunken spots on ripe fruit.					
Septoria Leaf Spot Small, gray circular leaf spots with dark borders.					
Fusarium Wilt Yellowing and wilting of foliage. Brown discoloration inside of stem.					Crop rotation, fumigation and resistant varieties are treatments. Maintain pH between 6.5 and 7.0.

Pest	Fungicide	Suggested Rate/Acre	PHI days	Maximum Use/Acre/Season	Remarks & Precautions
TOMATO, FIELD					
Buckeye Fruit Rot Circular, zonate bands within large spot on fruit, worse on lower clusters.	Gavel 75DF	1.5 - 2 lb	5	8 app	All are foliarly applied. Certain mefenoxam and metalaxyl products can be applied in drip irrigation, but have 28 day PHIs..
	Quadris 2.08F	5 - 6.2 fl oz	0	37 fl oz	
	Quadris Opti				
	5.5SC	1.6 pt	0	5 app	
	Presidio 4SC				
	Ridomil Gold	3-4 fl oz	2	12 fl oz	
Gray Mold (Botrytis) Gray, fuzzy mold on blighted foliage and fruits. Early Blight (See above)	Bravo				Endura may only be applied 2 times per season when applied at the gray mold rate. It is labeled for early blight control at 2.5 - 3.5 oz/A. Scala must be tank mixed with another one of these. Switch use only 2 times in a roll.
	Ridomil Gold	2.5 pt	14	3 app	
	Copper			3 app	
		2 lb	14		
	chlorothalonil 6L	2.75 pt	0	20 pt	
	Endura 70WG	9 - 12.5 oz	0	25 oz	
Leaf Mold Yellow spots on upper surface of leaf, olive to gray mold on underside.	Scala 5SC				Tanos must be tank mixed with and alternated with a non-strobilurin fungicide such as chlorothalonil or mancozeb. .
	Switch 62.5 WG	7 fl oz	1	35 oz	
		11-14 oz	0	56 oz	
	chlorothalonil 6L	2.75 pt	0	20 pt	
	mancozeb 80WP	1.5 - 3 lb	5	22.4 lb	
	Tanos 50WG	8 oz	3	72 oz	
Pythium Root and Stem Rot Dark, watery rot of lower stem of young plants.	Quadris Top	8 fl oz	0	47 fl oz	Apply MetaStar, Ridomil or Ultra Flourish preplant incorporated or see label for other methods. Aliette/Linebacker are foliar sprays. Apply Previcur via transplant water, drip, sprinkler, or direct nozzle to lower portion of plant and soil.
	MetaStar 2E AG	2-4 qt	28	6 qt	
	Ridomil Gold 4SL	1 - 2 pt/treated acre	28	3 pt	
	Ultra Flourish	2 - 4 pt/treated acre	28	6 pt	
	Aliette 80WDG				
	Previcur Flex 6F	2.5 - 5 lb	14	20 lb	
		1.5 pt	5	7.5 pt	

Pest	Fungicide	Suggested Rate/Acre	PHI (days)	Maximum Use/Acre/Season	Remarks & Precautions
TOMATO, FIELD					
Late Blight Large, irregular spots on leaves; firm rot of fruit.	Acrobat 50WP	6.4 oz	4	5 app	Begin applications before onset of disease and repeat on a 5- to 10-day schedule. Use 5-7 day intervals during mild, wet weather or if late blight is present. Most of these products require alternation with late blight fungicides with a different mode of action. Tank mix with a protectant fungicide such as chlorothalonil or mancozeb.
	Forum 4.18F	6 fl oz	4	5 app	
	Curzate 60DF	3.2 - 5 oz	3	30 oz	
Early Blight (See above)	Gavel 75DF	1.5 - 2 lb	5	8 app	
	Previcur Flex 6F	.7 – 1.5 pt	5	7.5 pt	
	Reason 4.13F	5.5 - 8.2 fl oz	14	24.6 fl oz	
	Ranman	2.1-2.75 fl oz	0	6 app	
	Revus	5.5 – 8.2 fl oz	14	32 fl oz	
	Presidio 4 SC	3 - 4 fl oz	2	12 fl oz	
Sclerotinia Stem Rot (Timber Rot) Dry, brown rot on stem. Hard, black pea-like structures form inside stem.					The Endura tomato label does not include this disease; however, applied as for early blight control, Endura should provide suppression of Sclerotinia.
Southern Blight Plants wilt and die. White mold can often be seen on base of stem.					Turn soil deep to bury plant debris which might harbor disease organisms. Use Terraclor in setting water ½ pt per plant.. Rotate with sod crops.
Verticillium Wilt Yellowing of leaves; inside of main stem has tan discoloration.					Crop rotation, fumigation, and resistant varieties are treatments.

Pest	Fungicide	Suggested Rate/Acre	PHI (days)	Maximum Use/Acre/Season	Remarks & Precautions
TOMATO, FIELD					
Phytophthora Blight (<i>P. capsici</i>) Rot of roots and crown.	MetaStar 2E AG	2 -4 qt	28	6 qt	Plant on raised beds, improve field drainage, and do not plant wet areas. Apply MetaStar, Ridomil or Ultra Flourish preplant incorporated or see label for other methods. Fosphite: Apply at 2 - 4 week intervals (see label). Apply Ranman in transplant water or to the base of the plant at trans[planting. Can also be applied to the foliage, but must involve in a resistance management program(see label)
	Ridomil Gold 4SL	1 – 2 pt/treated acre	28	3 pt	
	Ultra Flourish	2 - 4 pt/treated acre	28	6 pt	
	phosphorous acid: Fosphite	Pre-plant root dip: 2 qt/100 gal Drip irrig.: 2 - 3 qt in at least 100 gal	0		
	Ranman	2.75 fl oz	0	6 app	

Insect Identification

Pictures taken from Cornell University, Rutgers University, A.F. Sherf, R. Providenti, and Grainger County Producers



Aphids



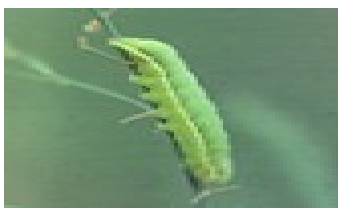
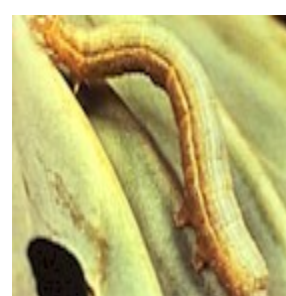
Spider Mites



White Flies

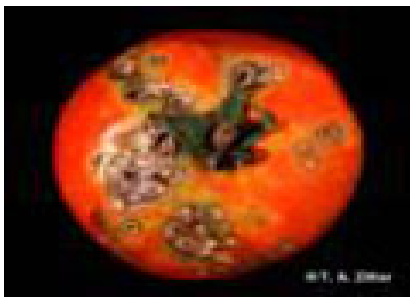


Cabbage Looper



Tomato Hornworm





Tarnish Plant Bug Injury



Stink Bug



Leaffooted Bug



Tobacco Hornworm



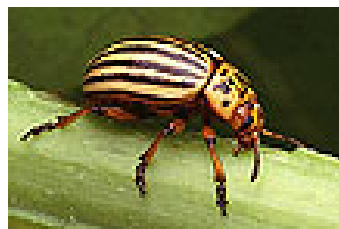
Tomato Fruitworm



Spotted Cucumber Beetle



Striped Cucumber Beetle



Colorado Potato Beetle



Blister Beetle

Greenhouse Tomato Insect Control

All tables taken from 2011 NC Agricultural Chemical Manual

Insect	Insecticide and Formulation	Amount of Formulation	Minimum Interval (Days) Between Last Application and Harvest	Precautions and Remarks
Aphid	endosulfan, MOA 2A (Thionex) 10 A 50 WP	1 lb/50,000 cu ft 1 lb/100 gal water	15 hr 2	Make sure the greenhouse is tightly closed, then apply in the air above the plants. The optimum temperature for application is 70 to 80°F. Keep greenhouse closed for at least 2 hr. Highly toxic—use with caution. Ventilate before reentry. Do not exceed five applications of endosulfan per year.
	3 EC	1 qt/100 gal water	2	
	imidacloprid, MOA 4A (Admire Pro) 4.6 F	0.6 fl oz/1,000 plants	0	Apply in a minimum of 21 gal water using soil drenches, micro-irrigation, or drip irrigation. Do not apply to immature plants as phytotoxicity may occur. Make only one application per crop per season. Will also control whiteflies.
	malathion, MOA 1B (various) 10 A 57 EC 25 WP	1 lb/50,000 cu ft 1 qt/100 gal water 4 lb/100 gal water	15 hr 1 1	
	insecticidal soap (M-Pede) 49 EC	2 tbsp/gal water	0	May be used alone or in combination. Acts as an exciter.
	<i>Beauveria bassiana</i> (Mycotrol WP)	0.25 lb/20 gal water	0	Apply when whiteflies are observed. Repeat in 4- to 5-day intervals.
Armyworm	endosulfan, MOA 2A (Thionex) 10 A 50 WP 3 EC	1 lb/50,000 cu ft 1 lb/50,000 cu ft 1 qt/100 gal water	15 hr 2 2	See remarks under Aphids (above). Do not exceed five applications of endosulfan per year.
	malathion, MOA 1B (various) 10 A 57 EC 25 WP	1 lb/50,000 cu ft 1 qt/100 gal water 4 qt/100 gal water	15 hr 1 1	See instructions for Aphids (above). Hazardous to honey bees.
	<i>Bacillus thuringiensis</i> (Javelin) WG (Agree) WP	0.5 lb to 1.25 lb/100 gal water 1 to 2 lb	0	
Cabbage looper	<i>Bacillus thuringiensis</i> , MOA 11 (various) (Javelin) WG (Dipel) DF	0.5 to 1 lb or 2 to 3 pt/100 gal water 0.5 to 1.25 lb/100 gal water 1 to 2 lb	0	
Climbing cutworm	See Armyworm			
Leafminer	malathion, MOA 1B (various) 10 A	1 lb/50,000 cu ft	15 hr	See TOMATO—Aphid.
	diazinon, MOA 1B (Diazinon, Spectracide) (AG 500) 50 WP	4 to 8 oz/100 gal water	3	Keep ventilators closed for 2 hr or overnight. Plant injury may result if labeling directions are not followed. For use by members of N.C. Greenhouse Vegetable Growers Association only.
Millipede, Cricket	malathion, MOA 1B (various) 5 D	Follow label directions		Apply to soil at base of plants. Do not contaminate fruit.
Pinworm	<i>Bacillus thuringiensis</i> , MOA 11 (Dipel) DF or (Javelin) WG	0.5 to 1 lb/100 gal water	0	
Slug	metaldehyde (various) bait	Follow label directions		Apply to soil surface around plants. Do not contaminate fruit.
Spider mite	naled (Dibrom) 60 EC	5 fl oz/50,000 cu ft OR 1 pt/100 gal water	1	See TOMATO — Aphid. Use predatory mites.
	insecticidal soap (M-Pede) 49 EC	2 tbsp/gal water	0	
Thrips	<i>Beauveria bassiana</i> (Mycotrol WP)	0.25 lb / 20 gal water	0	Use screens on intake vents. Apply when whiteflies observed. Repeat in 4- to 5-day intervals.
Tomato fruitworm	See Armyworm			

Greenhouse Tomato Insect Control (continued)

Insect	Insecticide and Formulation	Amount of Formulation	Minimum Interval (Days) Between Last Application and Harvest	Precautions and Remarks
Whitefly	malathionm MOA 1B (various) 10 A 57 EC 25 WP	1 lb/50,000 cu ft 1 qt/100 gal water 4 lb/100 gal water	15 hr 1 1	Apply as needed in the closed greenhouse in the air above the plants. Spray when the temperature is 70 to 80°F. Sprays every second day may be needed. Keep ventilator closed for 2 hr. Hazardous to honey bees.
	imidacloprid, MOA 4A (Admire Pro) 4.6 F	0.6 fl oz/1,000 plants	0	Apply in a minimum of 21 gal water using soil drenches, micro-irrigation, or drip irrigation. Do not apply to immature plants as phytotoxicity may occur. Make only one application per crop per season. Will also control aphids.
	insecticidal soap (M-Pede) 40 EC	2 tbsp/gal water	0	
	pyrethrins and PBO, MOA 3 (Pyrenone)	12 oz/ 20 gal water	0	May be used alone or tank mixed with a companion insecticide. (See label for details.)
	<i>Beauveria bassiana</i> (Mycotrol WP)	0.25 lb/ 20 gal water	0	Apply when whiteflies are observed. Repeat in 4- to 5-day intervals.

Field Insect Control

All tables taken from Southeastern U.S. 2011 Vegetable Crop Handbook

Pest	Insecticide	Suggested rate/acre	PHI (days)	Remarks & Precautions
Aphids & Flea Beetle	Dimethoate (Digon 400) 4EC	.5 - 1 pt.	7	Do not exceed rate with dimethoate as leaf injury may result
	Admire 2F	16 – 24 fl. ozs.	21	For short-term protection at planting. Admire Pro may also be applied to transplants in the planthouse not more than 7 days before planting at the rate of 0.44 (4.6 F formulation) or 1 oz (2 F formulation) per 10,000 plants. In the field, Admire Pro may be applied as an in-furrow spray directed on or below seed, a narrow surface band followed by irrigation, as a transplant drench, or through drip irrigation system.
	Admire Pro 4.6F	7 - 10.5 fl ozs.	21	
	Provado 1.6F	3.75 fl. ozs.	21	
	Fulfill 50WDG (Aphids only)	2.75 oz	0	
	Beleaf 50SG (will not control flea beetles)	2 – 2.8	0	
	Actara 25WDG	2 – 3 oz	0	
	Assail 30 SG	2 – 4 oz	7	
	Movento 2SC	4 – 5 fl oz	1	
	Platinum 2 SC	5 – 11 oz	30	Platinum may be applied to direct-seeded crops in-furrow seeding or transplant depth, post seeding or transplant as a drench, or through drip irrigation. Do not exceed 11 oz per acre per season of Platinum. Check label for plant-back restrictions for a number of crops.

Pest	Insecticide	Suggested rate/acre	PHI (days)	Remarks & Precautions
Cabbage Looper	Dipel DF	.5 – 1 lb	0	
	Crymax WDG	.5 – 1.5 lb	0	
&	Bifenthrin	2.6 – 6.4 fl oz	1	
	Baythroid XL (16.8 max use)	1.6 – 2.8 fl oz	0	
Hornworm	Coregen 1.67SC (foliar or drip)	3.5 – 5 fl oz	1	
&	Proclaim 5WDG	2.4 – 4.8 oz	7	
	Asana XL	4.8 – 9.6 oz	1	
Tomato Fruitworm (corn earworm)	<u>Danitol 2.4EC</u>	10.66 fl oz	3	<u>Do not exceed 42.6 oz/acre</u>
&	Synapse	2 – 3 oz	1	
	Proaxis	2.6 – 3.8 fl oz	5	
&	<u>Avaunt 30WDG</u>	2.5 – 3.5 oz	3	<u>No more than 14 oz/acre 5 days between sprays</u>
Pinworm	Warrior	2.6 – 3.8 oz	5	<u>Max 2.88 pt/acre don't use on cherry tomatoes.</u>
	Lannate 2.4 LV	1.5 – 3 pt	1	
	<u>Intrepid 2F</u>	4 – 10 oz	1	<u>Pinworm only low rate early</u>
	<u>Rimon</u>	9 – 12 fl oz	1	<u>3 app only max.</u>
		5 – 10 oz	1	
	Radiant	2.24 – 4 oz	1	
	Mustang Max			
Cutworms	Warrior	2.6 – 3.8 oz	5	Max 2.88 pt/acre don't use on cherry tomatoes
	Asana XL	4.8 – 9.6 oz	1	
	Baythroid XL	1.6 – 2.8 fl oz	0	Max 16.8 fl oz/acre
	Proaxis	2.6-3.8 fl oz	5	
	Mustang Max	3.2 – 4 oz	1	
Wireworm	Diazinon	2 – 4 qt		Broadcast and incorporate

Pest	Insecticide	Suggested rate/acre	PHI (days)	Remarks & Precautions
Colorado potato beetle	Admire 2F	16 fl. ozs.	21	Soil application for Admire and Admire Pro. Foliar applications use Provado.
	Admire Pro 4.6F	7 fl ozs.	21	
	Provado 1.6F	3.75 fl. ozs.	0	
	Assail 30 SG	1.5 – 2.5 oz	7	Foliar or drip
	Coragen	3.5 – 5 fl oz	1	
	Radiant	5 – 10 oz	1	Foliar only
	Actara	2 – 3 oz	0	
	Platinum 2 SC	5 – 11 oz	30	Platinum may be applied to direct-seeded crops in-furrow seeding or transplant depth, post seeding or transplant as a drench, or through drip irrigation. Do not exceed 11 oz per acre per season of Platinum. Check label for plant-back restrictions for a number of crops.
Leafminer	Agri-Mek	8 -16 fl oz	7	Max 48 fl oz/acre
	Trigard	2.66 oz	0	See label plant back
	Radiant	6 – 8 fl oz	1	Max 29 fl oz/acre
Spider Mite	Agri-mek	8 – 16 fl oz	7	Max 48 fl oz/acre
	Acramite	.75 – 1 lb	3	1 app only
	Portal	2 pt	3	2 app only
	Oberon	7 – 8.5 fl oz	7	3 app only
Stink Bug	Bifenthrin	2.6 – 6.4 fl oz	1	Max 42.6 fl oz/acre
	Danitol	10.66 fl oz	3	
	Proaxis	2.6 – 3.8 fl oz	5	
	Warrior	2.6 – 3.8 oz	5	
	Actara	3 – 5.5 oz	0	Max 11 oz/acre
	Mustang Max	3.2 – 4 oz	1	
Thrips	Dimethoate	.5 – 1 pt	7	See whitefly comments on Venom.
	Venom	1 – 4 oz foliar	1	
		5 – 6 oz soil	21	
	Monitor	1.5 – 2 pt	7	On foliage as needed. 3 app max Control on foliage not flower.
	Lannate	1.5 – 3 pt	1	
	Rimon	9 – 12 fl oz	1	
	Radiant	6 – 10 oz	1	

Pest	Insecticide	Suggested rate/acre	PHI (days)	Remarks & Precautions
Whitefly	Assail	2.5 – 4 oz	7	7 day intervals 5 app max
	Courier	9 – 13.6 fl oz	7	28 d interval 2 app max
	Venom	1 – 4 oz foliare	1	Do not follow soil applications with applications of other neonicotinoid insecticides (Actara, Provado, or Venom). Use only one application method. Soil applications may be applied in a narrow band on the plant row in bedding operation, as a post-seeding or transplant drench, as a side-dress after planting and incorporated 1 or more in., or through a drip irrigation system. May also control stink bugs.
		5 – 6 oz soil	21	
	Admire Pro	16 – 24 fl oz	21	
	Provado	7 – 10.5 fl oz	0	
	Knack	8 – 10 oz	14	
	Obenron	7 – 8.5 fl oz	7	
	Movento	4 – 5 fl oz	1	
	Platinum	5 – 11 oz	30	
	Actara	3 – 5.5 oz	0	

Pest	Insecticide	Suggested rate/acre	PHI (days)	Remarks & Precautions
Armyworm	Crymax	.5 – 1.5 lb	0	5-7 d interval when larva is small (for first 3 products)
	Dipel	.5 – 1 lb	0	
	Xentan	.5 – 1 lb	0	
	Coregent	3.5 – 4 fl oz	1	Foliar or drip
	Proclaim	2.4 – 4.8 oz	7	
	Synapse	2 – 3 oz	1	Max 14 oz/a 5 d intervals Low rate early Max 3 app
	Avaunt	3.5 oz	3	
	Intrepid	4 – 10 oz	1	
	Rimon	9 – 12 fl oz	1	
	Radiant	5 – 10 oz	1	

Varieties

Varieties for Greenhouses

Recommended for Bag Culture

‘Trust’
‘Match’
‘Switch’
‘Blitz’

Recommended for Ground Planting

Determinate
‘Celebrity’
‘Empire’
‘Emperador’
Indeterminate
‘Big Beef’

Varieties for Field Production

Local Sales Varieties

‘Red Delight’
‘Red Pride’
‘RFT 2734’ (‘Indy’)

Truck Cropping Varieties

‘Celebrity’
‘Empire’
‘Emperador’
‘Joker’
‘Fabulous’
‘Carolina Gold’
‘Big Beef’

Shipping Varieties

‘Amelia’
‘Floralina’
‘Florida 47’
‘Mountain Fresh’
‘Mountain Spring’
‘Sebring’

Spray Schedules

Each schedule is different. They are meant to be used as guides. The amount of money and severity of disease pressure will determine which schedule needs to be used. If you have any questions, ask your local Extension Agent.

Tomato Fungicides (Actigard) and Insecticides

Weeks after Transplanting	Fungicide	Rate (product/A)	PHI (days)	Cost (\$/A)	Insecticide	Rate (product/A)	PHI (days)	Cost (\$/A)
0	Dithane Rainshield 75DF	2 lb/A	5	\$6	Baythroid 2EC ^A	2.8 fl oz/A	0	\$8
1	Dithane Rainshield 75DF Actigard	2 lb/A 0.33 oz/A	5 14	\$6 \$18	Dimethoate 4EC ^A	1 pint/A	7	\$5
2	Dithane Rainshield 75DF Actigard	2 lb/A 0.33 oz/A	5 14	\$6 \$18	Baythroid 2EC ^A	2.8 fl oz/A	0	\$8
3	Dithane Rainshield 75DF Actigard	2 lb/A 0.5 oz/A	5 14	\$6 \$27	Dimethoate 4EC ^A	1 pint/A	7	\$5
4	Cabrio Actigard	8 oz/A 0.5 oz/A	1 14	\$12 \$27	Baythroid 2EC ^A	2.8 fl oz/A	0	\$8
5	Dithane Rainshield 75DF Actigard	3 lb/A 0.75 oz/A	5 14	\$9 \$41	Platinum (drip) Dimethoate 4EC	5 fl oz/A 1 pint/A	30 7	\$50 \$5
6	Cabrio Actigard	8 oz/A 0.75 oz/A	1 14	\$12 \$41	Danitol 2.4EC	10.67 fl oz/A	3	\$13
7	Dithane Rainshield 75DF	3 lb/A	5	\$9	Thiodan 3EC Xentari	1 quart/A 0.5 - 2 lb/A	2 0	\$9 \$6 - 24
8	Cabrio	12 oz/A	0	\$18	Danitol 2.4EC	10.67 fl oz/A	3	\$13

Tomato Disease and Insect Control Manual with Variety Selection

9	Bravo Weatherstik 6FL	3 pints/A	0	\$18	Thiodan 3EC DiPel DF	1 quart/A 1 - 2 lb/A	2 0	\$9 \$8 - 16
10	Cabrio	12 oz/A	0	\$18	Danitol 2.4EC	10.67 fl oz/A	3	\$13
11	Bravo Weatherstik 6FL	3 pints/A	0	\$18	Thiodan 3EC Xentari	1 quart/A 0.5 - 2 lb/A	2 0	\$9 \$6 - 24
12	Bravo Weatherstik 6FL	3 pints/A	0	\$18	Baythroid 2EC	2.8 fl oz/A	0	\$8
13	Bravo Weatherstik 6FL	3 pints/A	0	\$18	Thiodan 3EC DiPel DF	1 quart/A 1 - 2 lb/A	2 0	\$9 \$8 - 16
14	Bravo Weatherstik 6FL	3 pints/A	0	\$18	Baythroid 2EC	2.8 fl oz/A	0	\$8
15	Bravo Weatherstik 6FL	3 pints/A	0	\$18	Thiodan 3EC Xentari	1 quart/A 0.5 - 2 lb/A	2 0	\$9 \$6 - 24
16	Bravo Weatherstik 6FL	3 pints/A	0	\$18	Baythroid 2EC	2.8 fl oz/A	0	\$8

^A If Admire 2F is applied to transplants at 1 to 2.8 (2) oz/1,000 plants (\$48/A) or 1 pint/A (\$73/A), or if Platinum is applied @ 5 - 8 oz/A (\$50 – 80/A) these treatments may be omitted.

If mites become a problem, apply Kelthane 50WSP (2 day PHI) @ 1.25 lb/A at a cost of \$17/A.

If whiteflies become a problem, apply Knack 0.86EC (14 day PHI) @ 8 to 10 fl oz/A at a cost of \$31 - 39/A.

Tomato Fungicides (Tanos and Actigard) and Insecticides

Weeks after Transplanting	Fungicide	Rate (product/A)	PHI (days)	Cost (\$/A)	Insecticide	Rate (product/A)	PHI (days)	Cost (\$/A)
0	Dithane Rainshield 75DF	2 lb/A	5	\$6	Baythroid 2EC ^A	2.8 fl oz/A	0	\$8
1	Tanos Dithane Rainshield 75DF Cuprofix	8 oz/A 3 lb/A 2.5 lb/A	3 5 0	\$13 \$9 \$5	Dimethoate 4EC ^A	1 pint/A	7	\$5
2	Dithane Rainshield 75DF Actigard	2 lb/A 0.33 oz/A	5 14	\$6 \$18	Baythroid 2EC ^A	2.8 fl oz/A	0	\$8
3	Tanos Dithane Rainshield 75DF Cuprofix	8 oz/A 3 lb/A 2.5 lb/A	3 5 0	\$13 \$9 \$5	Dimethoate 4EC ^A	1 pint/A	7	\$5
4	Dithane Rainshield 75DF Actigard	2 lb/A 0.5 oz/A	5 14	\$6 \$27	Baythroid 2EC ^A	2.8 fl oz/A	0	\$8
5	Tanos Dithane Rainshield 75DF Cuprofix	8 oz/A 3 lb/A 2.5 lb/A	3 5 0	\$13 \$9 \$5	Platinum (drip) Dimethoate 4EC	5 fl oz/A 1 pint/A	30 7	\$50 \$5
6	Dithane Rainshield 75DF Actigard	2 lb/A 0.75 oz/A	5 14	\$6 \$41	Danitol 2.4EC	10.67 fl oz/A	3	\$13
7	Tanos Dithane Rainshield 75DF Cuprofix	8 oz/A 3 lb/A 2.5 lb/A	3 5 0	\$13 \$9 \$5	Thiodan 3EC Xentari	1 quart/A 0.5 - 2 lb/A	2 0	\$9 \$6 - 24
8	Dithane Rainshield 75DF Actigard	2 lb/A 0.75 oz/A	5 14	\$6 \$41	Danitol 2.4EC	10.67 fl oz/A	3	\$13

Tomato Disease and Insect Control Manual with Variety Selection

9	Bravo Weatherstik 6FL Cuprofix	3 pints/A 2.5 lb/A	0 0	\$18 \$5	Thiodan 3EC DiPel DF	1 quart/A 1 - 2 lb/A	2 0	\$9 \$8 - 16
10	Bravo Weatherstik 6FL	3 pints/A	0	\$18	Danitol 2.4EC	10.67 fl oz/A	3	\$13
11	Bravo Weatherstik 6FL Cuprofix	3 pints/A 2.5 lb/A	0 0	\$18 \$5	Thiodan 3EC Xentari	1 quart/A 0.5 - 2 lb/A	2 0	\$9 \$6 - 24
12	Bravo Weatherstik 6FL	3 pints/A	0	\$18	Baythroid 2EC	2.8 fl oz/A	0	\$8
13	Bravo Weatherstik 6FL Cuprofix	3 pints/A 2.5 lb/A	0 0	\$18 \$5	Thiodan 3EC DiPel DF	1 quart/A 1 - 2 lb/A	2 0	\$9 \$8 - 16
14	Bravo Weatherstik 6FL	3 pints/A	0	\$18	Baythroid 2EC	2.8 fl oz/A	0	\$8
15	Bravo Weatherstik 6FL Cuprofix	3 pints/A 2.5 lb/A	0 0	\$18 \$5	Thiodan 3EC Xentari	1 quart/A 0.5 - 2 lb/A	2 0	\$9 \$6 - 24
16	Bravo Weatherstik 6FL	3 pints/A	0	\$18	Baythroid 2EC	2.8 fl oz/A	0	\$8

^A If Admire 2F is applied to transplants at 1 to 2.8 (2) oz/1,000 plants (\$48/A) or 1 pint/A (\$73/A), or if Platinum is applied @ 5 - 8 oz/A (\$50 - 80/A) these treatments may be omitted.

If mites become a problem, apply Kelthane 50WSP (2 day PHI) @ 1.25 lb/A at a cost of \$17/A.

If whiteflies become a problem, apply Knack 0.86EC (14 day PHI) @ 8 to 10 fl oz/A at a cost of \$31 - 39/A

Tomato Irrigation and Fertigation Schedule

Weeks after Transplanting	Irrigation Rate (inches/A/day) ^a	Irrigation Rate (gallons/A/week) ^b	N Rate (lb N/A/day) ^c	Fertilizer Product	Fertilizer Rate (lb/A/week) ^d
Preplant				12-24-24 ^e	500 ^e
0	0.10 - 0.15	6,300 - 9,500			
1	0.10 - 0.15	6,300 - 9,500			
2	0.10 - 0.15	6,300 - 9,500			
3	0.10 - 0.15	6,300 - 9,500			
4	0.20 - 0.25	12,600 - 15,800	1	Calcium Nitrate	50
5	0.20 - 0.25	12,600 - 15,800	1	Potassium Nitrate	50
6	0.20 - 0.25	12,600 - 15,800	1	Calcium Nitrate	50
7	0.25 - 0.30	15,800 - 19,100	1	Potassium Nitrate	50
8	0.25 - 0.30	15,800 - 19,100	1	20-20-20	50
9	0.25 - 0.30	15,800 - 19,100	1	Calcium Nitrate	50
10	0.25 - 0.30	15,800 - 19,100	1	Potassium Nitrate	50
11	0.25 - 0.30	15,800 - 19,100	1	20-20-20	50
12	0.25 - 0.30	15,800 - 19,100	1	Potassium Nitrate	50
13	0.20 - 0.25	12,600 - 15,800	1	20-20-20	50
14	0.20 - 0.25	12,600 - 15,800	1	Potassium Nitrate	50
15	0.20 - 0.25	12,600 - 15,800	1	20-20-20	50
16	0.20 - 0.25	12,600 - 15,800	1	Potassium Nitrate	50

^a The water requirement of a plant increases as the plants become larger. In addition, as temperatures increase,

- more water is lost through transpiration. After the peak of production, the watering rate may be reduced.
- ^b The gallons of water/A/week is based on 27,156 gallons of water/A inch and 2 ft wide beds on 6 ft centers. Therefore, only 1/3 of the soil area is receiving water (2 ft out of 6 ft). If 2 ft beds were spaced on 5 ft centers, then the gallons required would be 15% higher.
- ^c This is a broadcast rate (the entire acreage, not just the area under the plastic). If calculating on a treated acre (the acre under the plastic), the rate would be 2.5 - 3 lb N/A/day.
- ^d This is a broadcast rate (the entire acreage, not just the area under the plastic). This rate can be applied once a week or split over several applications.
- ^e If a soil test is generated, then apply 60 lb/A N and P₂O₅ and K₂O according to the soil test recommendations. However, if no soil test is generated, then apply a fertilizer that results in 50 - 60 lb/A N and 100 - 120 lb/A of each P₂O₅ and K₂O (50-100-100 to 60-120-120).

Tomato Herbicides

Time of Application	Herbicide	Rate (product/A)	PHI (days)	Cost (\$/A)
Burndown	Gramoxone Max 3SL Non-ionic Surfactant (NIS)	1 qt/A 0.25% (v/v) (8 oz)	30	\$10 \$1
	or Roundup Weathermax 5.5	1 - 2 qt/A		\$15 - 30
Pretransplant or Post-transplant ^A	Sencor 75DF	0.5 - 0.67 lb/A	7	\$10 - 13
	Sencor 75DF Select 2EC COC	0.5 - 0.67 lb/A 8 fl oz/A 1% (v/v) (32oz)	7 20	\$10 - 13 \$11 \$3
Directed ^A Post-transplant	Sencor 75DF Select 2EC COC	0.5 - 0.67 lb/A 8 fl oz/A 1% (v/v) (32 oz)	7 20	\$9 - 13 \$11 \$3
	or Sencor 75DF Gramoxone Max 3SL Non-ionic Surfactant	0.5 - 0.67 lb/A 1 qt/A 0.25% (v/v) (8 oz)	7 30	\$10 - 13 \$10 \$1

^A Poast 1.5EC (20 day PHI) can be applied @ 1.5 pints/A at a cost of \$13/A. Use COC @ 1% v/v.

Admire Treatment of Tomato Transplants

1. Mix 2 to 2.8 fl oz of Admire in 1 quart of water.
2. Treat 1,000 tomato plants with the 1 quart of water.
3. Rinse the plants with another quart of water.

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